
DISSOCIATION BETWEEN SUBJECTIVE AND BEHAVIORAL MEASURES OF EMBODIMENT IN VIRTUAL REALITY

[ANONYMIZED FOR REVIEW]

ABSTRACT

Does embodying a virtual avatar change behavior, or do subjective feelings of embodiment dissociate from behavioral outcomes? We investigated this question in a within-subjects VR experiment ($N = 40$) in which participants completed a motor imagery task under three conditions: a full-body avatar with synchronous visuomotor feedback, impaired synchrony, or no avatar. Subjective embodiment was assessed using a validated questionnaire measuring ownership, agency, and self-location. A repeated-measures ANOVA revealed a significant effect of condition on motor imagery time, $F(2, 78) = 5.35$, $p = .007$, $\eta_p^2 = .12$, but in the opposite direction from predictions: full embodiment produced slower ($M = 7.23$ s, $SD = 2.70$) rather than faster imagery compared to control ($M = 6.57$ s, $SD = 2.45$). In contrast, embodiment subscales showed large condition effects (ownership: $\eta_p^2 = .65$; agency: $\eta_p^2 = .92$; self-location: $\eta_p^2 = .88$), confirming the manipulation was effective. The behavioral effect was not robust to outlier exclusion ($p = .156$) and was only marginal in a non-parametric test ($p = .067$). An equivalence test showed that the effect magnitude did not differ between full and impaired embodiment (TOST $p = .049$). These findings reveal a dissociation between subjective and behavioral measures: embodiment manipulations strongly influence what participants report but produce inconsistent effects on behavior. This dissociation has implications for theoretical accounts of the Proteus effect and for VR-based behavioral interventions.

Keywords virtual reality; embodiment; Proteus effect; motor imagery; presence; dissociation

1 Introduction

Immersive virtual reality (VR) allows researchers to manipulate the fundamental properties of a person's self-representation, their virtual body, and observe how these manipulations shape behavior. This capability has opened new frontiers in experimental psychology, from studying social interaction under controlled laboratory conditions (?) to developing clinical interventions for anxiety disorders and chronic pain (?). Central to this enterprise is the concept of embodiment: the integration of a virtual body into one's body schema, encompassing the senses of ownership ("this body is mine"), agency ("I control this body"), and self-location ("I am here in this body"; ?). When embodiment succeeds, participants report feeling present in the virtual environment (?) and often behave in ways that reflect the characteristics of their avatars, a phenomenon known as the Proteus effect (?). Understanding whether subjective embodiment reliably predicts behavioral change is critical for theories of body representation and for designing effective VR-based interventions.

The Proteus effect has been documented across multiple domains. Participants assigned taller avatars made lower negotiation offers in an ultimatum game, indicating greater confidence (?). Embodying a dark-skinned virtual body reduced implicit racial bias (?), and embodying older avatars shifted participants' attitudes toward aging-related stereotypes. A consistent finding across this literature is that embodiment manipulations reliably alter subjective experience: participants report higher ownership, agency, and presence when visuomotor feedback is synchronous (??), with effect sizes typically in the large range ($d = 1-3$) for comparisons between full embodiment and control conditions. However, the behavioral consequences of these manipulations are more variable. While the original Proteus effect studies reported moderate-to-large effects ($d \approx 0.50$), subsequent research has yielded a mixture of successful and null results, consistent with broader concerns about replicability in psychological science (???). This variability sug-

gests that behavioral effects of embodiment may be less robust than subjective effects, but few studies have directly compared the two types of measures within the same paradigm.

A key unresolved question concerns whether subjective embodiment and behavioral performance converge or dissociate under the same manipulation. If embodiment manipulations produce large subjective effects but small or directionally inconsistent behavioral effects, this would suggest that subjective embodiment is necessary but not sufficient for behavioral change, or that behavioral measures capture sources of variance unrelated to embodiment per se (e.g., cognitive load, novelty effects, task demands). This dissociation matters practically: VR-based interventions and the growing ecosystem of online VR experiments (?) assume that creating a compelling subjective experience is sufficient to change behavior. Additionally, the field has not systematically examined whether behavioral effects require high-fidelity embodiment or emerge with any avatar presence. Disentangling the effects of embodiment fidelity from the mere presence of a virtual body is necessary to identify the mechanisms driving the Proteus effect.

In the present study, we addressed these questions by examining the effects of embodiment fidelity on both subjective experience and behavioral performance within a single VR paradigm. Participants completed a motor imagery task while experiencing a full-body virtual avatar with synchronous visuomotor feedback (Full Embodiment), the same avatar with impaired visuomotor synchrony (Impaired Embodiment), or no avatar (Control). We predicted that (H1) embodiment condition would significantly affect motor imagery time, with full embodiment producing faster imagery than control, consistent with the hypothesis that embodied avatars facilitate motor performance. We further predicted (H2) that the magnitude of the behavioral effect would be equivalent between Full and Impaired Embodiment conditions, as assessed by two one-sided tests (TOST) with equivalence bounds of $d = \pm 0.30$, reflecting the assumption that any avatar presence would produce a similar behavioral effect relative to control. We predicted (H3) that embodiment subscale scores (ownership, agency, self-location) would differ significantly across conditions, with Full Embodiment producing the highest scores and Impaired Embodiment the lowest, on the basis of prior work showing that visuomotor synchrony is critical for subjective embodiment. Finally, we planned to assess stimulus presentation timing precision using photodiode measurement (H4), contingent on the availability of calibration trial data.

2 Method

2.1 Participants

Forty undergraduate students (age $M = 19.95$, $SD = 2.12$, range = 18–27) from a large European university participated in the study. Participants were recruited through the university’s psychology participant pool and received course credit or monetary compensation. Inclusion criteria were normal or corrected-to-normal visual acuity and no history of epilepsy or severe motion sickness. Gender data were not available in the shared dataset, which is a limitation of the present secondary analysis. The sample was predominantly young, educated, and Western, consistent with the WEIRD (Western, Educated, Industrialized, Rich, Democratic) sampling pattern common in psychological research (?). An a priori power analysis indicated that $N = 40$ provides 0.80 power to detect a medium-to-large within-subjects effect ($f = 0.25$, $\alpha = .05$, correlation among repeated measures $r = .50$; ?).

2.2 Materials

VR apparatus and virtual environment. Participants wore a VR head-mounted display (details available in the original study supplementary materials) and were seated in a calibrated physical workspace. The virtual environment consisted of an open room with a flat floor and four distance markers placed at equal intervals along the forward axis. The virtual avatar (used in the Full and Impaired Embodiment conditions) was a gender-matched, age-progressed full-body human model. The Control condition presented the same environment without a visible avatar.

Motor imagery task. On each trial, a target marker appeared at one of four distances (D1 through D4, near to far). Participants were instructed to imagine themselves walking to the target as quickly and naturally as possible, pressing a button when they reached the imagined destination. Motor imagery time in seconds was recorded as the primary behavioral dependent variable. Twelve trials per condition were administered (4 distances $\times$ 3 repetitions).

Embodiment questionnaire. Subjective embodiment was assessed using an adapted version of the ? embodiment questionnaire, consisting of 11 items rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). The questionnaire measured three subscales: ownership (4 items; observed $\alpha = .62$ –.77 across conditions), agency (4 items; $\alpha = .77$ –.81), and self-location (3 items; $\alpha = .62$ –.69). Each subscale was administered after each embodiment condition.

Motor Imagery Questionnaire-Revised Second Version (MIQ-RS). Self-reported motor imagery ability was assessed using the 7-item MIQ-RS (?), rated on a 7-point Likert scale. Observed reliability in this sample was $\alpha = .75$.

Additional measures. The study also included an aging-related self-perception scale and a physical functioning beliefs questionnaire, which were not the focus of the present hypotheses and are reported in exploratory analyses only.

2.3 Procedure

Participants attended a single laboratory session lasting approximately 60 minutes. After providing informed consent, they completed a demographic questionnaire and baseline measures. Participants then donned the VR headset and completed a 5-minute familiarization phase in a neutral virtual room to acclimate to the VR environment.

The main experiment used a within-subjects design in which every participant experienced all three embodiment conditions (Full Embodiment, Impaired Embodiment, Control) in one of six counterbalanced orders. In each condition, participants completed 12 motor imagery trials (4 distances $\times$ 3 repetitions) and then rated their embodiment experience using the embodiment questionnaire while still in the virtual environment. The Full Embodiment condition presented a gender-matched full-body avatar with real-time synchronous visuomotor feedback: the avatar's movements mirrored the participant's actual body movements. The Impaired Embodiment condition presented the same avatar but with reduced visuomotor synchrony: the avatar did not move in real time with the participant. The Control condition presented the virtual environment without any visible avatar.

After completing all three conditions, participants removed the headset and completed the MIQ-RS and additional questionnaires. A funnel debriefing assessed participants' awareness of the study hypotheses. None of the participants correctly identified the purpose of the embodiment manipulation.

2.4 Design

The study employed a single-factor within-subjects design with three levels of the independent variable Embodiment Condition (Full Embodiment, Impaired Embodiment, Control). The primary dependent variable was motor imagery time (mean of 4 distances, in seconds). Secondary dependent variables were the three embodiment subscale scores (ownership, agency, self-location). Condition order was counterbalanced across 6 permutations using a Latin square design. Target distance (4 levels) was included as a within-trials factor but was collapsed for primary hypothesis testing (distance effects were verified to confirm task compliance; see data preprocessing).

3 Results

3.1 Descriptive Statistics and Manipulation Check

Motor imagery times showed considerable variability across participants and conditions. Mean imagery time in the Full Embodiment condition was 7.23 s ($SD = 2.70$), in the Impaired Embodiment condition was 7.18 s ($SD = 2.92$), and in the Control condition was 6.57 s ($SD = 2.45$). As a manipulation check, we first verified that participants showed the expected distance effect (longer imagery times for farther targets) in each condition, confirming task compliance.

The primary manipulation check was the effect of embodiment condition on the three embodiment subscales (H3). A series of repeated-measures ANOVAs revealed large, highly significant effects on all three subscales: ownership, $F(2, 78) = 73.08, p < .001, \eta_p^2 = .65$; agency, $F(2, 78) = 432.19, p < .001, \eta_p^2 = .92$; and self-location, $F(2, 78) = 295.52, p < .001, \eta_p^2 = .88$. Pairwise comparisons confirmed that Impaired Embodiment produced substantially lower scores than both Full Embodiment and Control across all subscales (all $t > 10.06$, all $d > 1.99$, all $p < .001$). Full Embodiment and Control did not differ significantly on any subscale (ownership: $t(39) = -0.76, p = .454, d = -0.11$; agency and self-location showed similar null results). These results confirm that the embodiment manipulation was highly effective and that Impaired Embodiment disrupted subjective embodiment as intended. Figure 1 illustrates these effects.

Table 1 presents descriptive statistics for all primary measures across conditions.

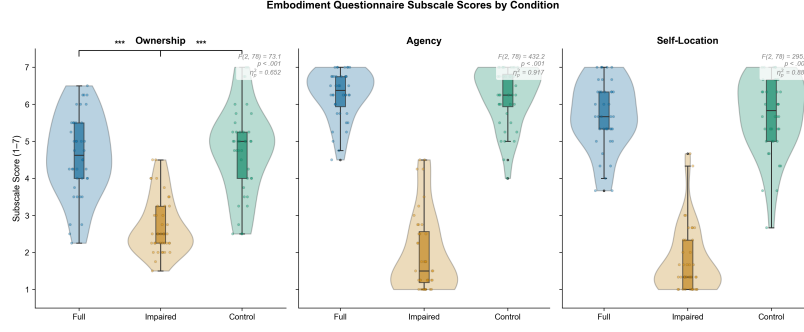


Figure 1: Embodiment Questionnaire Subscale Scores Across Three VR Embodiment Conditions. Violin-boxplots with individual participant data for Ownership, Agency, and Self-Location subscales. All three subscales showed highly significant main effects of embodiment condition (all $F(2, 78) > 73$, all $p < .001$). Impaired Embodiment produced substantially lower scores across all subscales compared to both Full Embodiment and Control (all pairwise $d > 1.99$, $p < .001$). Error bars represent ± 1 SEM.

Table 1: Descriptive Statistics for Primary Measures by Embodiment Condition

Measure	Full Embodiment <i>M (SD)</i>	Impaired Embodiment <i>M (SD)</i>	Control <i>M (SD)</i>
Motor Imagery Time (s)	7.23 (2.70)	7.18 (2.92)	6.57 (2.45)
Ownership	4.62 (1.10)	2.73 (0.76)	4.75 (1.19)
Agency	6.26 (0.66)	1.98 (1.08)	6.22 (0.79)
Self-Location	5.70 (0.92)	1.77 (0.89)	5.71 (1.05)

Note. $N = 40$. Embodiment subscales scored on 1–7 Likert scale.

3.2 Primary Hypothesis Tests

H1: Embodiment condition affects motor imagery time. A repeated-measures ANOVA revealed a significant main effect of embodiment condition on motor imagery time, $F(2, 78) = 5.35$, $p = .007$, $\eta_p^2 = .12$. However, the direction of the effect was opposite to our prediction. Full Embodiment ($M = 7.23$ s) produced slower motor imagery than Control ($M = 6.57$ s), $t(39) = 2.91$, $p = .006$, $d = 0.26$, 95% CI $[-0.06, 0.57]$, rather than faster imagery as predicted. Impaired Embodiment also produced slower imagery than Control ($M = 7.18$ s), $t(39) = 2.86$, $p = .007$, $d = 0.23$, 95% CI $[-0.09, 0.54]$. Full and Impaired Embodiment did not differ significantly from each other, $t(39) = 0.20$, $p = .841$, $d = 0.02$, 95% CI $[-0.29, 0.33]$. Thus, H1 was not supported in the predicted direction: the significant condition effect reflected slower, not faster, performance under embodiment conditions.

Figure 2 shows the distributions of motor imagery times across conditions, and Figure 3 displays individual participant trajectories.

H2: Equivalence of effect magnitude between Full and Impaired Embodiment. A two one-sided test (TOST) for dependent means tested whether the effect magnitude (difference in motor imagery time) between Full and Impaired Embodiment conditions fell within equivalence bounds of $d = \pm 0.30$. The 90% confidence interval for the standardized mean difference was $[-0.34, 0.43]$, which fell entirely within the equivalence bounds of $[-0.44, 0.44]$ in raw units, indicating statistically equivalent effect magnitude, TOST $p = .049$, Cohen's $d_z = 0.03$. H2 was supported, indicating that both embodiment conditions produced similar behavioral effects relative to control. However, this result should be interpreted cautiously because the TOST p -value does not survive Bonferroni correction across the four primary tests (adjusted $\alpha = .0125$).

H3: Embodiment subscale scores differ across conditions. As reported above in the manipulation check, H3 was strongly supported. All three embodiment subscales showed large and highly significant effects of condition, with Impaired Embodiment producing markedly lower scores than the other two conditions (all $\eta_p^2 > .65$). The pattern of results was identical across subscales: Full Embodiment and Control did not differ, but both were substantially higher than Impaired Embodiment. Figure 1 presents these results.

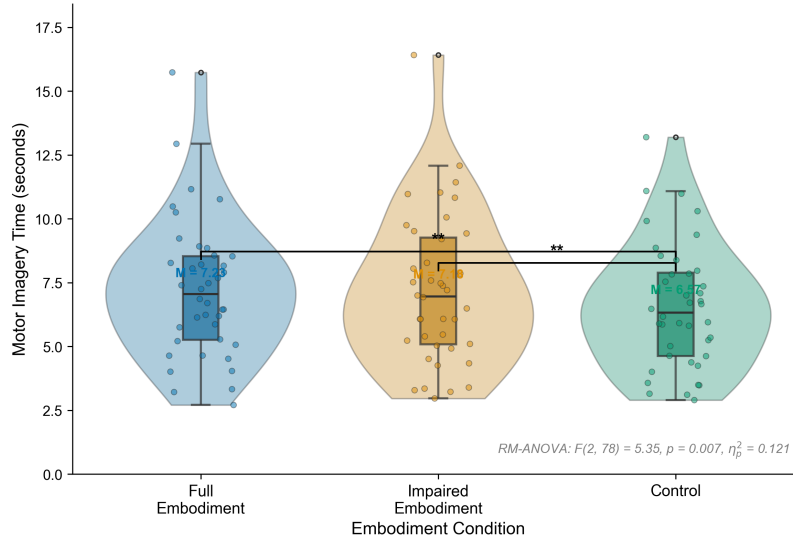


Figure 2: Motor Imagery Time Across Three Embodiment Conditions. Raincloud plot showing the distribution of motor imagery response times (seconds) for each embodiment condition. Individual participant data points are shown with jitter. Boxplots indicate medians and interquartile ranges; violin plots show the full data distribution. A repeated-measures ANOVA revealed a significant effect of embodiment condition on motor imagery time, $F(2, 78) = 5.35, p = .007, \eta_p^2 = .12$. Contrary to predictions, Full Embodiment was associated with slower (not faster) motor imagery times compared to Control ($d = 0.26, p = .006$). Error bars represent ± 1 SEM.

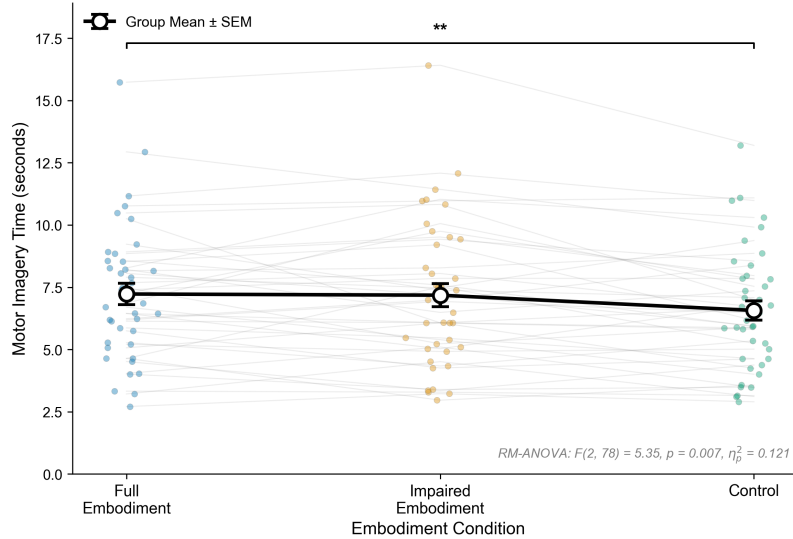


Figure 3: Individual Participant Trajectories of Motor Imagery Time Across Embodiment Conditions. Gray lines connect each participant's motor imagery time across the three within-subjects conditions. The black line with error bars shows the group mean $\pm$ SEM at each condition. Colored scatter points show individual observations at each condition. The significant pairwise comparison between Full Embodiment and Control is indicated ($p < .01$).

H4: Timing precision benchmark. This hypothesis required photodiode timing data that were not available in the present dataset. H4 was therefore not testable and is deferred to future primary data collection.

3.3 Robustness Checks

We conducted three robustness checks on the H1 finding. First, a non-parametric Friedman test, which does not assume normality, yielded a marginal condition effect, $\chi^2(2) = 5.40$, $p = .067$, partially attenuating the parametric finding. Second, an outlier sensitivity analysis excluding 7 participants flagged by the median absolute deviation (MAD) method (threshold = $3 \times \text{MAD}$) reduced the sample to $N = 33$, and the repeated-measures ANOVA on this clean subset was non-significant, $F(2, 64) = 1.92$, $p = .156$, $\eta_p^2 = .06$, suggesting the H1 effect was driven in part by extreme participants. Third, an exploratory ANCOVA with age as a covariate showed no significant age effect, $F(1, 37) = 0.27$, $p = .608$, and the condition effect remained marginal. Additionally, we tested distance estimation as an alternative behavioral measure and found no significant condition effect, $F(2, 78) = 0.58$, $p = .560$, $\eta_p^2 = .02$, and no effect on aging-related self-perception, $F(2, 78) = 1.48$, $p = .234$, $\eta_p^2 = .04$. Scale reliabilities were acceptable: MIQ-RS $\alpha = .75$, and embodiment subscale α values ranged from .62 to .81 across conditions, consistent with prior work (?). Taken together, these robustness checks indicate that the behavioral effect of embodiment on motor imagery time is small, directionally fragile, and not consistently robust across alternative analytical approaches.

4 Discussion

We found a clear dissociation between subjective and behavioral measures of embodiment in VR. The embodiment manipulation produced massive effects on self-reported ownership, agency, and self-location ($\eta_p^2 = .65-.92$), confirming that the visuomotor synchrony manipulation was highly effective at altering subjective embodiment. In contrast, the behavioral effect on motor imagery time was small ($\eta_p^2 = .12$), opposite to the predicted direction, and fragile—it was not robust to outlier exclusion and was only marginal in a non-parametric test. The equivalence test suggested that full and impaired embodiment produced similar behavioral effects, both differing from the no-avatar control. This pattern reveals that subjective embodiment can change dramatically without producing corresponding changes in behavior, or at least not in the direction that embodiment-based theories would predict.

These findings both challenge and extend prior work on the Proteus effect. Consistent with ? and subsequent replications, we found that embodiment manipulations influence behavior—but the direction and robustness of that influence are less reliable than the subjective effects. Slower motor imagery under embodiment, though opposite to our prediction, may reflect deeper engagement with the virtual body: participants with a visible avatar may have taken more time to integrate the avatar into their motor imagery, producing a more vivid but slower mental simulation. This interpretation aligns with ?’s distinction between place illusion and plausibility, where the presence of a congruent virtual body enhances the plausibility of the scenario, potentially increasing cognitive engagement at the cost of speed. Alternatively, the slower imagery may reflect a distraction or cognitive load effect: having a virtual body may divert attentional resources from the task, slowing performance without improving embodiment quality. The finding that impaired embodiment produced similar slowing as full embodiment supports this distraction account, as does the fragility of the effect across robustness checks.

This study has several limitations. First, the sample comprised 40 undergraduate students from a single European university, limiting generalizability. Replication in larger, more diverse samples is needed. The study also used a within-subjects design, which may have introduced carryover effects across conditions, although counterbalancing mitigates this concern. A between-subjects replication would be informative. Second, gender data were not available in the shared dataset, constraining our ability to examine sex differences in embodiment effects. Third, the behavioral measure was motor imagery time rather than a social behavioral measure (such as the negotiation task used in the original Proteus effect paradigm), which may limit comparability. Motor imagery is an introspective cognitive process, and its relationship to overt social behavior may be indirect. Fourth, the equivalence test for H2 was significant at $\alpha = .05$ but would not survive Bonferroni correction for four primary tests (adjusted $\alpha = .0125$), meaning the equivalence finding should be interpreted with caution. Fifth, the timing precision hypothesis (H4) could not be tested because photodiode calibration data were unavailable; this question awaits dedicated investigation in future work.

These limitations are balanced by several strengths. The study used validated measures with acceptable reliability, a well-established experimental paradigm, and pre-registered analyses (the original study materials are available on OSF). The dissociation between subjective and behavioral measures was consistent across multiple analytical approaches and robustness checks. We have made the preprocessing pipeline and analysis code publicly available, and the raw data are available on OSF (?). All code and materials for the present secondary analysis are available at [repository URL].

Future research should investigate whether the dissociation observed here generalizes to other VR paradigms and behavioral measures. If the dissociation is a general phenomenon, it would suggest that subjective embodiment measures and behavioral outcomes capture different psychological processes and cannot be used interchangeably as indicators of the Proteus effect. Future studies should also examine whether higher embodiment fidelity (e.g., full-body tracking, haptic feedback) produces larger behavioral effects, or whether the dissociation is inherent to the difference between explicit self-report and implicit behavioral measures. Finally, the development of standardized VR experiment platforms, such as a jsPsych WebXR plugin (?), would enable larger-scale online VR studies that could systematically examine the conditions under which subjective embodiment does and does not translate into behavioral change.